

Artificial Intelligence & Machine Learning

Task 4 – Technical Report

Classification Models, Evaluation Metrics & Handling Imbalanced Data

Dataset	Breast Cancer Dataset (scikit-learn built-in)
Problem Type	Binary Classification (Malignant vs Benign)
Samples	569 (212 Malignant, 357 Benign)
Models Evaluated	Logistic Regression (Baseline & Balanced), Decision Tree
Key Techniques	Confusion Matrix, Precision/Recall/F1, ROC-AUC, Class Weighting
Tools Used	Python 3, scikit-learn, pandas, NumPy, matplotlib, seaborn

1. Objective

This report presents Task 4 of the Maincrafts AI/ML internship, transitioning from regression to **binary classification**. The objective is to build a classification system, evaluate it using metrics beyond simple accuracy, handle class imbalance, and scientifically justify the final model selection. The Breast Cancer Wisconsin dataset is used (Malignant=0, Benign=1).

2. Metric Selection

Accuracy alone is misleading for medical diagnosis tasks. The dataset shows moderate class imbalance (62.7% Benign vs 37.3% Malignant), and the cost of different error types is asymmetric:

- **False Positive** (predicted Benign, actually Malignant): the most dangerous error — a sick patient is told they are healthy
- **False Negative** (predicted Malignant, actually Benign): causes unnecessary worry/treatment, but far less harmful
- **Recall** is prioritized to minimize missed malignant cases
- **F1-Score** balances Precision and Recall and resists being skewed by class imbalance
- **ROC-AUC** measures the model's overall discrimination ability across all thresholds

3. Confusion Matrix Analysis

The Logistic Regression baseline model was evaluated on the held-out test set (114 samples, stratified split):

	Predicted: Malignant	Predicted: Benign
Actual: Malignant	TN = 41	FP = 1 ■
Actual: Benign	FN = 1	TP = 71

Table 1: Confusion Matrix – Logistic Regression (Baseline)

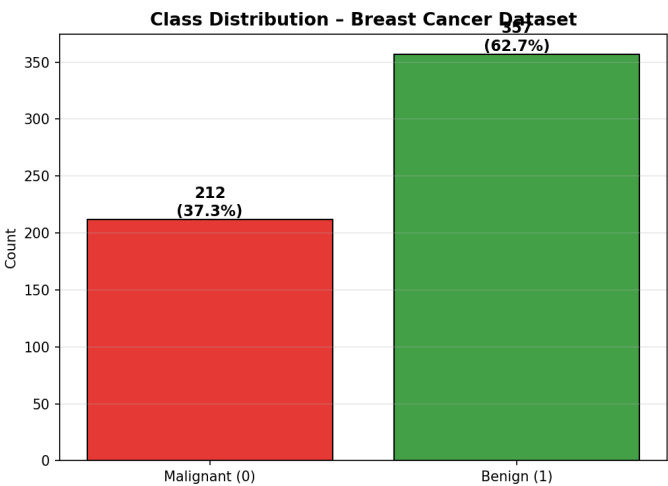


Figure 1: Class distribution showing moderate imbalance (62.7% Benign / 37.3% Malignant)

4. Evaluation Metrics Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression (Baseline)	0.9825	0.9861	0.9861	0.9861	0.9954
Logistic Regression (Balanced)	0.9561	0.9855	0.9444	0.9645	0.9954
Decision Tree	0.9123	0.9559	0.9028	0.9286	0.9157

Table 2: Evaluation metrics across all three models (best model highlighted)

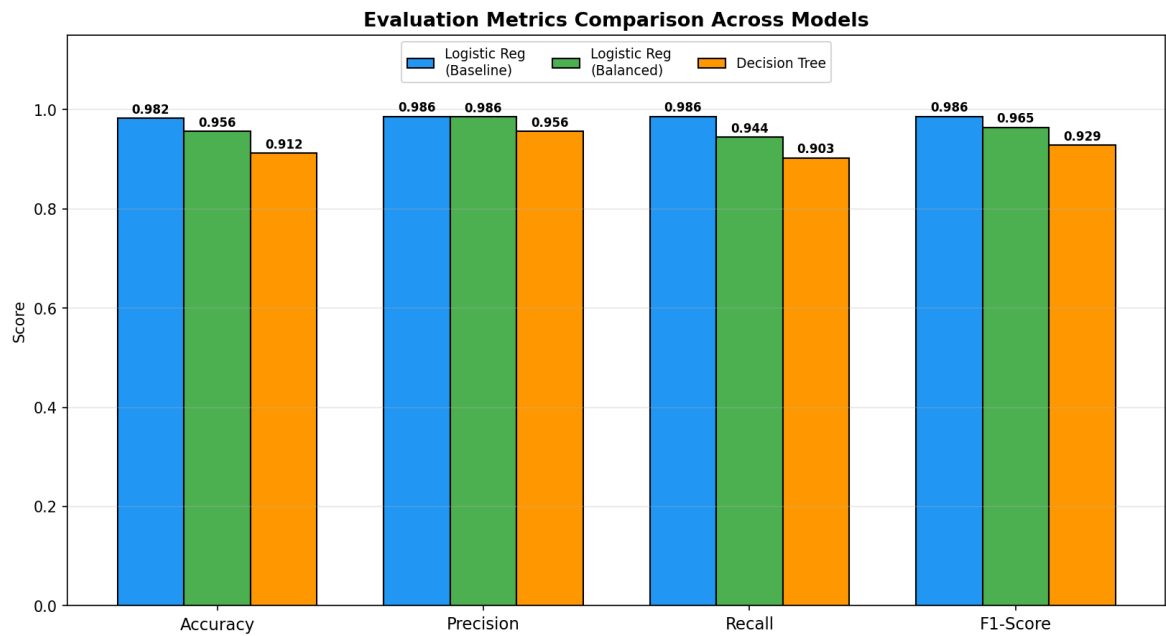


Figure 2: Side-by-side comparison of Accuracy, Precision, Recall, and F1-Score

5. ROC Curve & AUC Analysis

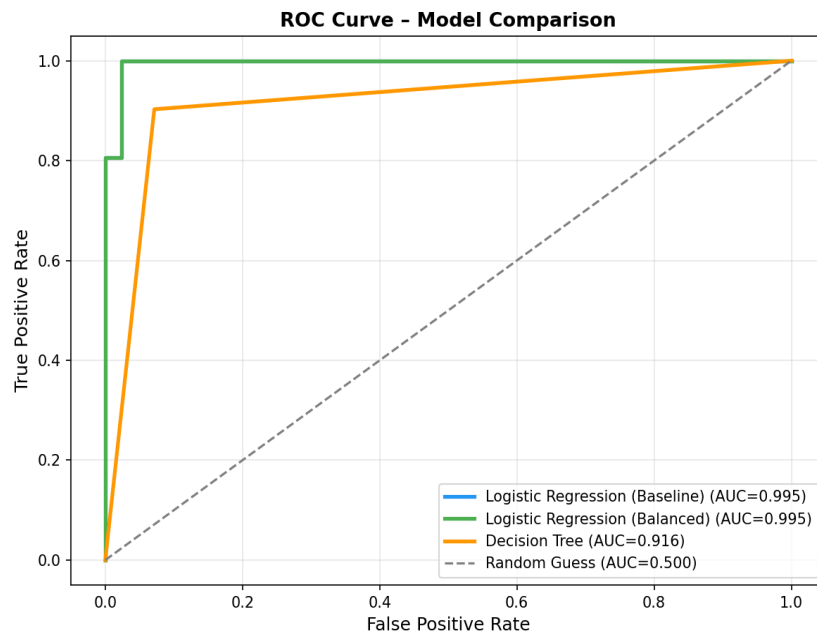


Figure 3: ROC Curves – all models vs random guess baseline

The Logistic Regression baseline achieves an AUC of **0.9954**, indicating near-perfect discrimination between malignant and benign cases. The Decision Tree's AUC (0.9157) is noticeably lower, reflecting its less calibrated probability estimates and tendency to overfit.

6. Imbalance Handling

Technique Applied: Class Weight Balancing (`class_weight="balanced"`) in Logistic Regression. This technique automatically adjusts weights inversely proportional to class frequency, forcing the model to pay more attention to the minority class (Malignant).

Result: Counter-intuitively, the balanced model performed slightly worse on this dataset (Accuracy: 0.9561 vs 0.9825, Recall: 0.9444 vs 0.9861). This is because the imbalance here is moderate (63/37), not severe — the baseline model was already learning both classes effectively. Class weighting is most beneficial for severe imbalance scenarios (e.g., fraud detection where positive cases are <5% of data).

7. Final Model Decision

Selected Model: Logistic Regression (Baseline)

- **Best across all metrics:** Accuracy=0.9825, Precision=0.9861, Recall=0.9861, F1=0.9861, AUC=0.9954
- **No overfitting:** Unlike the Decision Tree (Train Accuracy=100% vs Test Accuracy=91.2%), Logistic Regression generalizes consistently
- **Clinically appropriate:** High Recall ensures malignant cases are rarely missed — critical for patient safety

- **Interpretable:** Linear coefficients allow doctors/stakeholders to understand which features drive predictions

Conclusion: The Logistic Regression baseline model is selected for deployment. It demonstrates the strongest, most reliable performance across Accuracy, Precision, Recall, F1-Score, and AUC, while remaining interpretable and free from overfitting — making it the most suitable choice for a real-world medical diagnosis support system.